TIC 355809352

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

```
BLS EPHEMERIS
  Period       14.719197 d  (ALIASED, see below)
    -> corrected period: 29.438395 d
  Epoch (t0)   2944.1295 BTJD
  Duration     0.0840 d
  Depth        0.077 %  (773 ppm)
  SNR          51.8

INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     3  (gate >= 3)
  Robust depth ratio    0.96  (gate >= 0.50)
  Robust depth sigma    16.1  (gate >= 7.0)
  Odd/even z-score      3.39  (ALIAS FLAGGED)
  Secondary eclipse     not detected
```

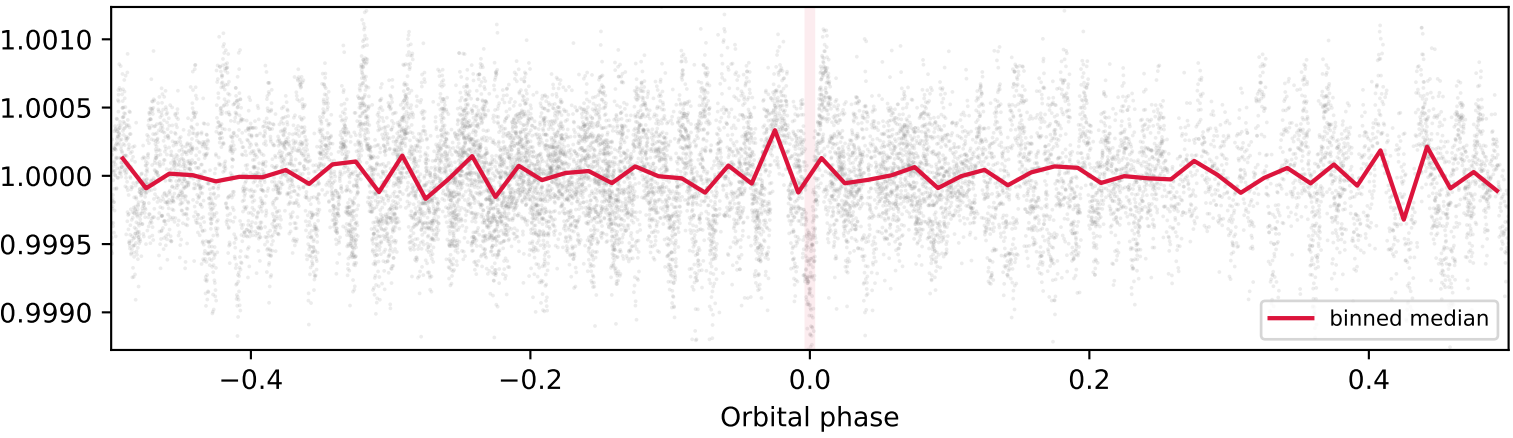
```
CATALOG CROSS-MATCH (novelty)

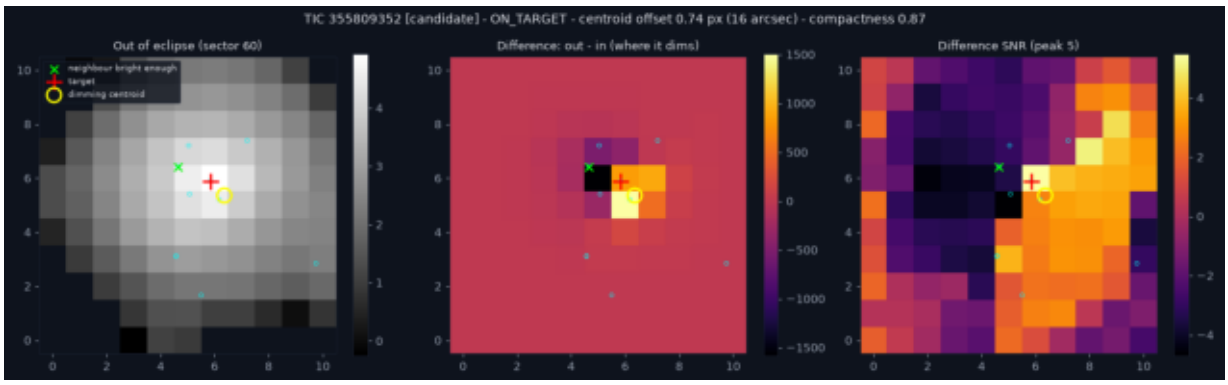
  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 60    light-curve dip in this sector: 7.5 sigma    t0 refined by 0.3600 phase
  Difference-image peak SNR   5.48   (gate >= 5.0)      Compactness   0.87   (gate >= 0.50)
  Dimming centroid offset     0.74 px (16 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 355809351
    2.02 px away, dTmag +4.64 - farther from the dimming than the target
```

Phase-folded light curve (shaded = BLS eclipse window)





Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 3.39 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 29.438395 d, in any write-up.